CYBER SECURITY
TASK 1

VULNERABILITY ASSESSMENT REPORT

Website Tested: https://testphp.vulnweb.com
Tools Used: Nmap and OWASP ZAP

Prepared By: ***DURGA R***

durga ramasamy

Summary:

- This task was done to check the security of a demo website.
- The goal was to find common security problems that could make the website unsafe.
- Nmap was used to scan the website and find open ports and server information.
- OWASP ZAP was used to detect security vulnerabilities in the web application.
- The scan showed some security issues, such as:
- Missing security headers
- Weak or improper server configuration
- These issues could allow attackers to exploit the website if they are not fixed.
- By applying proper security settings and best practices, the website's overall security can be improved.

Tools Used:

1. Nmap

Nmap is a network scanning tool used to discover open ports and gather information about the server.

2. OWASP ZAP

OWASP ZAP is a security testing tool used to detect vulnerabilities in web applications.

3. Web Browser

A browser was used to manually observe website behavior and verify results.

Vulnerability Scan Results:

The following screenshots show the results of the vulnerability assessment performed on the demo website using Nmap and OWASP ZAP.

Nmap Scan Result

- The Nmap scan was used to detect open ports and server information.
- The result shows port 80 (HTTP) open on the target website.
- The server is running nginx version 1.19.0, which reveals server information.

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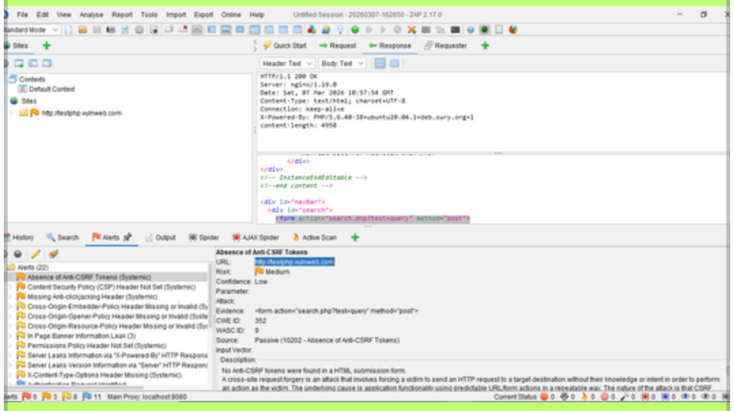
C:\Users\Lenovo>nmap -sV testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-07 21:35 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.32s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.19.0

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 53.48 seconds
```

Figure 1: Nmap scan showing open port 80 and the nginx 1.19.0 server version.

OWASP ZAP Scan Result

- OWASP ZAP identified multiple security issues in the web application.
- The alerts include absence of Anti-CSRF tokens, missing security headers, and server information disclosure.
- These vulnerabilities indicate that the website lacks important security protections.



OWASP ZAP alert results showing identified vulnerabilities.

Security Findings:

1. Missing Anti-Clickjacking Header
The website does not include the X-Frame-Options header, which may allow clickjacking attacks.

2. Absence of Anti-CSRF Tokens
Forms on the website do not contain CSRF tokens, which could allow unauthorized actions.

3. Server Version Disclosure
The Nmap scan revealed the server version (nginx 1.19.0), which may help attackers identify potential vulnerabilities.

Security Recommendations:

1. A vulnerability assessment was performed on the demo website using Nmap and OWASP ZAP.
2. The scan identified several security weaknesses such as missing security headers, absence of Anti-CSRF tokens, and server information disclosure.
3. These vulnerabilities could potentially be exploited by attackers if proper security measures are not implemented.
4. Implementing recommended security practices such as adding security headers, hiding server information, and using HTTPS can significantly improve the website's security.
5. Regular security testing and monitoring are important to maintain a safe and secure web environment.

Conclusion:

- The vulnerability assessment identified several security weaknesses in the website. Although these issues are common, they could potentially be exploited if left unaddressed.
- By applying recommended security measures and performing regular assessments, the overall security posture of the website can be significantly improved.